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 Studierichting: Informatica
 Oefeningenreeks 3D nr. 2.8

$$f(x) = \frac{x^2}{x-1}$$

Domein: $\text{dom } f = \mathbb{R} \setminus \{1\}$

Afgeleide:

$$f'(x) = \frac{2x(x-1) - x^2 \cdot 1}{(x-1)^2} = \frac{2x^2 - 2x - x^2}{(x-1)^2} = \frac{x^2 - 2x}{(x-1)^2} = \frac{x(x-2)}{(x-1)^2}$$

Nulpunten:

$$x(x-2) = 0 \Leftrightarrow x = 0 \text{ of } x = 2$$

Verloopschema:

x	$-\infty$	0	2	$+\infty$	
$f'(x)$	+	0	-	0	+
$f(x)$	$\nearrow$	0	$\searrow$	4	$\nearrow$
		(max)		(min)	